Estimation of Daily All-Wave Surface Net Radiation With Multispectral and Multitemporal Observations From GOES-16 ABI

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***Abstract*— As a vital parameter describing the Earth surface energy budget, surface all-wave net radiation (R_n) drives many physical and biological processes. Remote estimation of R_n using satellite data is an effective approach to monitor the spatial and temporal dynamics of R_n. Accurate daily R_n estimation typically depends on the spatio-temporal resolutions of satellite data. There are currently few high-spatial-resolution daily R_n products from polar-orbiting satellite data, and they exhibit limited accuracy due to sparse diurnal observations. In addition, traditional estimation approaches typically require cloud mask and clear-sky albedo as inputs and ignore the length ratio of daytime (LRD), which may lead to large errors. To overcome these challenges and obtain R_n data with improved spatial resolution and accuracy, an operational approach was proposed in this study to derive daily 1-km R_n, which takes the advantages from a radiative transfer model, a machine learning algorithm, and multispectral and dense diurnal temporal information of geostationary satellite observations. An improved all-sky hybrid model (AHM) coupling radiative transfer simulations with a random forest (RF) model was first developed to estimate the shortwave net radiation (R_{ns}). Then, another RF model was developed to estimate the daily R_n from R_{ns}, incorporating the LRD, which is called extended hybrid model (EHM). Data from the Advanced Baseline Imager (ABI) onboard the new-generation Geostationary Operational Environmental Satellite (GOES)-16 with a 5-min temporal resolution and a 1-km spatial resolution were used to test the proposed method. Compared to traditional lookup table (LUT) algorithms, the results show that AHM not only makes the process of R_{ns} estimation simple and efficient but also has high accuracy in estimating instantaneous all-sky R_{ns}. Benefiting from high spatio-temporal resolutions, our daily R_{ns} estimates using GOSE-16 data exhibited superior performance compared to using the 1-km Moderate Resolution Imaging Spectroradiometer (MODIS) and 1° Clouds and the Earth's Radiant Energy System (CERES) product. Using accurate daily R_{ns} estimates and LRD as inputs, the EHM model shows reasonably good results for estimating R_n (R^2, RMSE, and bias of 0.91, 20.95 W/m^2, and −0.05 W/m^2, respectively). Maps of 1-km R_{ns} and R_n exhibit similar spatial patterns to those from the 1° CERES product, but with substantially more spatial details. Overall, the proposed R_n retrieval scheme can accurately estimate all-sky 1-km R_{ns} and R_n at mid- to low-latitudes and can be easily adapted and applied to other GOES-16-like satellites, such as Himawari-8, Meteosat Third Generation (MTG), and Fenyun-4. This study demonstrates the advantages of estimating R_n using geostationary satellites with improved accuracy and resolutions.**

***Index Terms*— All-sky hybrid model (AHM), daily net radiation, extended hybrid model (EHM), geostationary satellite, length ratio of daytime (LRD).**

I. Introduction

SURFACE all-wave net radiation (R_n) describes the Earth surface energy budget [1], [2]. It drives many physical and biological processes, such as evapotranspiration and heating the air and soil [3]–[5]. R_n is the difference between the total downwelling and upwelling radiation in the shortwave (0.3–3 μm) and longwave (3–100 μm) spectral domains. Mathematically, R_n is the sum of the shortwave net radiation (R_{ns}) and longwave net radiation (R_{nl}), which can be calculated according to the following equation:

$$R_n = R_{ns} + R_{nl}. \tag{1}$$

The R_n level is influenced by both atmospheric conditions and surface properties, such as the cloud and aerosol optical depth, surface albedo, surface temperature, and emissivity. Clouds and aerosols primarily affect downwelling radiation components, while surface albedo and temperature/emissivity greatly impact upwelling radiation components. R_n is a key parameter used to connect the atmosphere system with the near-surface and explore their energy exchanges [6]–[9]. Hence, obtaining reliable R_n datasets is crucial.

There are three data source types that can monitor R_n: ground-based measurements, reanalysis, and remote sensing products. Ground-based measurements provide temporally continuous R_n data with high accuracy (uncertainties are ~5 W/m^2 for pyrgeometers); however, their applications are restricted to limited spatial coverage due to sparsely distributed observation sites. Although R_n can also be obtained from reanalysis products with improved spatial coverage, the grid scales are coarse, ranging from approximately 0.25° to 0.625°, which limits their fine-scale usage [10]–[12]. Remote sensing offers tremendous advantages in estimating regional and global R_n [3], [13], and some existing remote sensing products (~1° spatial resolution) also provide R_n estimates, such as Clouds and the Earth's Radiant Energy System (CERES),

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Global Energy and Water Experiment (GEWEX), and International Satellite Cloud Climatology Project (ISCCP). The CERES product is considered more accurate than other remote sensing datasets [14], but it has typically been restricted to studying changes in the energy budget at the global scale due to its coarse spatial resolution of 1° [15], [16]. Few works focus on R_n estimation at the kilometer scale [10], [17]. So far, such high spatial resolution (1 km) R_n with high accuracy is still lacking, which motivates us to develop an operational approach for estimating R_n.

Satellite observations are typically used to estimate the instantaneous R_n at the time of satellite overpass, which does not represent the average status of R_n during the day [18]. However, climate change and modeling studies typically require daily R_n as the input [19]. As obtaining the daily R_n involves coupling surface and atmosphere information with large spatial and temporal variations, remote sensing data used to estimate R_n must exhibit both multispectral and multitemporal characteristics [20]. Previous studies usually used polar-orbiting satellite sensors, such as the Moderate Resolution Imaging Spectroradiometer (MODIS), to estimate R_n due to their high spatial resolution and multispectral bands [13], [21]–[23]. However, as the MODIS sensors onboard the Terra and Aqua satellites scan the mid- to low-latitudes approximately twice during the day, daily R_n estimates may exhibit limited accuracy, particularly for days with large variations in cloud optical properties [23]–[25]. Geostationary satellites provide great potential to map daily R_n with more diurnal observations compared with polar-orbiting satellites. However, previous-generation geostationary satellites were not equipped with multispectral bands. For example, Geostationary Operational Environmental Satellite (GOES)-12 through GOES-15 only contained one visible and two infrared bands with spatial resolutions of 1–4 km. Few studies used these satellites to derive R_n because the surface characteristics cannot be modeled well by their limited spectral bands [3]. With the development of remote sensors, new-generation geostationary satellites, such as GOES-16, Himawari-8, Meteosat Third Generation (MTG), and Fenyun-4, can provide remote sensing data with increased spectral bands and improved spatio-temporal resolutions compared to previous-generation geostationary satellites and polar-orbiting satellites [26], [27]. For example, GOES-16 contains 16 spectral bands, and it scans the continental United States (CONUS) every 5 min at a spatial resolution of 1–2 km [28]. Thus, the potential of estimating R_n from new-generation geostationary satellite data should be evaluated.

Many methods have been developed to remotely estimate R_n [10], [17], [29]–[35]. These methods can be roughly grouped into three categories: parameterization-based, hybrid model-based, and empirical methods. Parameterization methods derive R_n estimates by calculating and then summing the four radiation components [21], [23], [36]. Because parameterization methods depend heavily on various atmosphere and surface remote-sensing products, they may result in limited R_n retrieval accuracy due to the propagation of uncertainty [7], [23], [37], [38]. Traditional hybrid model methods, also known as lookup table (LUT) algorithms, utilize simplified online radiative transfer calculations to estimate R_n [35], [39]–[41]. However, few studies have applied this method to estimate cloudy-sky R_n because real-time and accurate cloud parameter products are not always available for characterizing the longwave radiation level [42]–[44]. R_n is strongly related to shortwave radiation, based on which empirical methods have been developed to estimate R_n by leveraging shortwave radiation, i.e., downwelling solar radiation (DSR) or R_{ns} [45]–[48]. Due to their simplicity and effectiveness, empirical methods may be more suitable for estimating R_n at the regional and global scales [48]–[52]. Additionally, Hu *et al.* [46] have demonstrated that R_{ns} contributes to more accurate R_{n} estimates compared to using DSR because it considers the influence of the surface albedo [50]. Previous studies typically used DSR as a main predictor of R_n estimation instead of R_{ns} because many DSR products are readily available [27], [53]. However, separately estimating DSR and albedo may lead to greater uncertainty than directly estimating R_{ns} from satellite top-of-atmosphere (TOA) observations [54]. In addition, there are missing values in remote sensing estimated surface albedo datasets under cloudy conditions [11]. Unlike previous studies, this study derives R_n from accurate all-sky R_{ns} estimates.

The length ratio of daytime (LRD), defined as the ratio of the daytime duration to the entire day, is a vital parameter for modeling daily R_n because the relationship between daily R_{ns} and daily R_n may vary with a change in the LRD [17]. However, previous studies ignored the effect of LRD on R_n estimation model [10], [48]. In this study, the impact of LRD on the relationship between R_{ns} and R_n was explored, and R_n retrieval algorithm included the variable LRD. Generally, hybrid methods are more convenient and flexible solutions for retrieving R_{ns} under all-sky conditions [1], [24], [41], [55]. Traditional hybrid methods typically require cloud mask products to distinguish the sky conditions before estimating R_{ns} [56]. However, the accuracy is often inconsistent between cloud mask products over different scenes [57], [58]. In this study, an all-sky hybrid model (AHM) coupling radiative transfer simulation with a machine learning algorithm is proposed to directly estimate all-sky R_{ns} without a cloud mask product.

In this study, we estimated 1-km daily R_n with multispectral and multitemporal observations from new-generation geostationary satellite data. We focused on generating operational algorithms for net radiation and used GOES-16 as the proxy data to: 1) develop and validate an AHM model to estimate all-sky R_{ns}; 2) improve daily R_{ns} estimates using multitemporal satellite observations; 3) improve daily R_n estimates by leveraging high-accuracy R_{ns} estimates; and 4) map high-resolution daily R_{ns}/R_n over the CONUS. To the best of our knowledge, this is the first study to estimate all-sky R_n from new-generation geostationary satellites. Moreover, the proposed estimation scheme takes the advantages from a radiative transfer model, a machine learning algorithm, and multispectral and dense diurnal temporal information from geostationary satellite observations, which can be further used to generate a daily 1-km R_n dataset.

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II. Data

GOES-16 Level 1B TOA data was used to estimate R_{ns}. To illustrate the advantage of GOES-16 in estimating daily R_{ns}, MODIS TOA data was also used to estimate daily R_{ns} for comparison. In addition to GOES-16-derived R_{ns}, other auxiliary variables, i.e., Modern-Era Retrospective analysis for Research and Applications version 2 (MERRA-2)-derived daily air temperature (Ta) and relative humidity (RH) and GLASS-derived fractional vegetation cover (FVC), were also included in this study to estimate R_n. Ground measurements were collected to evaluate the accuracy of the R_{ns} and R_n estimates. The radiation flux remote-sensing product CERES was used to examine and compare the spatial distributions of R_{ns} and R_n estimates from GOES-16 data over the CONUS.

A. GOES-16 Data

GOES-16, formerly known as GOES-R before the satellite reached its geostationary orbit, is the first satellite of the GOES-R series jointly operated by the National Aeronautics and Space Administration (NASA) and National Oceanic and Atmospheric Administration (NOAA). GOES-16 provides a remote view from space that is centered on the Americas. It became fully operational on December 18, 2017. The Advanced Baseline Imager (ABI) aboard the GOES-16 contains 16 spectral bands, including two visible bands, four near-infrared (NIR) bands, and ten infrared bands, and it observes the Earth surface at high spatio-temporal resolutions [59]. Its spatial resolution ranges from 0.5 to 2 km, and temporal resolution is 5 min for the CONUS and 15 min for the full disk region. The blue (Band 1), red (Band 2), NIR (Band 3), and shortwave-infrared (Band 5) bands were selected for R_{ns} estimation. Detailed descriptions of the four bands are provided in Table I. To match the spatial resolution of the selected bands, Band 2 was integrated to 1 km, and thus, the estimated R_{ns}/R_n exhibits a resolution of 1 km. In this study, the TOA spectral radiance product (Level 1B) of GOES-16 in 2018, which can be freely downloaded from the NOAA official website (https://www.ncdc.noaa.gov/data-access/satellite-data/goes-r-series-satellites), was used to compute the TOA reflectance. The radiometric calibration of GOES-16 ABI Level 1B product demonstrates high accuracy with an uncertainty of 2%–5% [59]. Satellite-solar viewing geometry is not provided officially. Therefore, the solar zenith angle (SZA), solar azimuth angle (SAA), viewing zenith angle (VZA), and viewing azimuth angle (VAA) must be precalculated. Given the geolocation information and date/time, the solar position can be easily computed. Additionally, we followed the guideline provided by the NOAA to derive the VZA/VAA for each GOSE-16 pixel (https://www.goes-r.gov/users/docs/PUG-L1b-vol3.pdf). SZA, VZA, and relative azimuth angle (RAA) were used in R_{ns} estimation model, wherein RAA is the relative angle between SAA and VAA.

Although our AHM approach does not require cloud information as the input, the ABI cloud mask product was still used to compare the accuracy of instantaneous R_{ns} estimates under clear- and cloudy-sky conditions. The ABI Level 2 clear sky mask product (ABI-L2-ACMC), which exhibits an accuracy of approximately 86.0% [60], was selected to distinguish the sky conditions.

TABLE I
Information of Selected GOES-16 Bands

Band No.	Type	Wavelength range (μm)	Central wavelength (μm)	Spatial resolution (km)
1	Blue	0.45–0.49	0.47	1
2	Red	0.59–0.69	0.64	0.5
3	NIR	0.846–0.885	0.865	1
5	SWIR	1.58–1.64	1.61	1

B. MODIS Data

The MODIS is a sensor onboard the Terra satellite, which was launched in 1999, and Aqua satellite, which was launched in 2002. There are about two daytime observations for most mid- to-low-latitudes when combining MODIS-Terra and MODIS-Aqua data, with overpass times of approximately 13:30 and 10:30 local solar time due to the orbit design. MODIS Level 1B version 6 products, including MOD021KM and MYD021KM, provide TOA observational data at a 1-km spatial resolution. The TOA reflectance at the first seven land bands (459–2155 nm) and band 19 (915–965 nm) derived from the MOD021KM and MYD021KM products were used to estimate R_{ns} with an AHM model. Corresponding geolocation information and viewing geometry data (e.g., SZA, VZA, and RAA) were obtained from the MOD03 and MYD03 products. Both Aqua data and Terra data were used to estimate daily R_{ns} because previous studies have demonstrated that R_{ns} estimates combining these two satellites exhibit higher accuracy than those using observations from a single MODIS sensor [24]. Daily R_{ns} estimates derived from the two MODIS sensors were compared with those from GOES-16 to illustrate the degree of improvement in R_{ns} retrieval as a result of using high-frequency observations from the new-generation geostationary satellite.

C. CERES Product

The CERES project offers satellite-based observations of the Earth radiation budget and cloud information since 2000. Measurements from CERES instruments onboard several satellites including the Aqua, Terra, and Suomi National Polar-orbiting Partnership and auxiliary information from many other instruments were combined to generate comprehensive radiation products for studying climate, weather, and applied sciences [61]. The CERES product was selected as a reference product for benchmarking the spatial patterns of R_{ns}/R_n because it typically provides more accurate radiation estimates than other existing satellite remote sensing products, such as ISCCP and GEWEX [62]. The CERES Level 3 Synoptic Radiative Fluxes and Clouds (SYN) product provides daily all-sky DSR, USR, DLR, and ULR estimates at a spatial

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resolution of 1°, allowing daily R_{ns} and R_n to be easily calculated. Surface radiation fluxes from CERES-SYN were derived using the Fu-Liou radiative transfer codes coupled with MODIS and geostationary satellite-derived cloud products, atmospheric profile data, and other auxiliary variables as the inputs [61], [63]. Additionally, the estimated surface radiation fluxes were constrained by the CERES TOA flux data. The CERES-SYN Edition 4.1 product over the CONUS was requested and downloaded from https://ceres.larc.nasa.gov/.

D. MERRA-2 Reanalysis Data

MERRA-2, which includes recent advances in modeling and data assimilation and is intended to replace the previous-generation MERRA dataset, is a suite of global atmospheric reanalysis products generated by the NASA Global Modeling and Assimilation Office [64]. MERRA-2 provides hourly global surface water vapor, Ta, and specific humidity data at a spatial resolution of $0.5° \times 0.625°$ from 1980 onward [12]. For R_{ns} retrieval, the MERRA-2 water vapor product was applied to the AHM for water vapor absorption correction. For R_n estimation, the retrieval model also required the Ta and RH as inputs. Although the RH is not directly provided by MERRA-2, it can be calculated using the specific humidity data of MERRA-2. The daily minimum, maximum, and mean Ta and daily mean RH in 2018 over the CONUS were derived and/or calculated from MERRA-2.

E. GLASS FVC Product

Surface vegetation influences the relationship between R_{ns} and R_n. Some studies have utilized the normalized difference vegetation index (NDVI) or enhanced vegetation index (EVI) to model this relationship [50], [65]. However, there are many missing values in the NDVI and EVI products. The FVC is highly correlated with the NDVI, and it describes the proportion of vegetation in the pixel. This study used the GLASS FVC instead of the NDVI or EVI due to its complete spatio-temporal coverage and higher accuracy [66]. To generate the GLASS FVC product, Jia *et al.* [66] developed a reliable and accurate FVC retrieval algorithm from MODIS surface reflectance data (RMSE = 0.157) according to a comparison against ground-based measurements. After performing gap-filling methods, such as spatio-temporal interpolation, the GLASS FVC was spatially continuous at the global scale, demonstrating great potential for various vegetation monitoring and applications [67]. The GLASS FVC product exhibits a 500-m spatial resolution and 8-d temporal resolution, and it can be freely downloaded from http://www.glass.umd.edu/Download.html.

F. Ground Measurements

Ground measurements provide accurate surface radiation observations that can be used to validate the performance of our proposed R_{ns}/R_n retrieval models. In this study, ground measurements were acquired from two reliable radiation flux observation networks, including Surface Radiation (SURFRAD) and AmeriFlux (https://ameriflux.lbl.gov/).

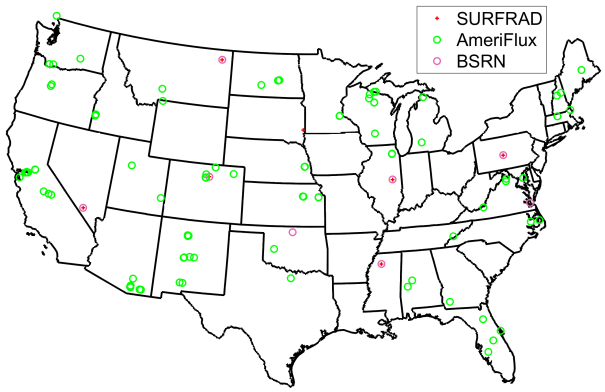


Fig. 1. Locations of sites at SURFRAD, AmeriFlux, and BSRN networks.

The SURFRAD network was established in 1993 and is supported by the NOAA to enhance our understanding of climate change and the surface energy budget by providing accurate, continuous, and long-term ground measurements of surface radiation fluxes in differing climatic areas over the CONUS. Since January 1, 2009, the SURFRAD data has been recorded at 1-min intervals; the previous temporal interval was 3 min. The instruments equipped at SURFRAD sites record both shortwave and longwave radiation components, e.g., DSR, upwelling solar radiation (USR), downwelling longwave radiation (DLR), and upwelling longwave radiation (ULR), and their respective data quality flags. To validate R_{ns} estimates from GOES-16, MODIS, and the reference product, R_{ns} data from seven SURFRAD sites in 2018 was used, which can be downloaded from http://www.esrl.noaa.gov/gmd/grad/surfrad/. To study the carbon dioxide, water, and energy exchanges between terrestrial ecosystems and the atmospheric system in North, Central, and South America, AmeriFlux was launched in 1996, and it is currently supported by the Lawrence Berkeley National Laboratory. Data from AmeriFlux sites is reported at 30-min time intervals and is available at http://ameriflux.lbl.gov/. Ground measurements from 97 AmeriFlux sites were also collected to develop and validate the R_n retrieval model. To support the World Climate Research Programme (WCRP), the Baseline Surface Radiation Network (BSRN) was established to provide high quality ground radiation fluxes data at 1-min intervals. To perform additional evaluation of the R_n retrieval model, ground measurements from eight BSRN sites were also collected. Fig. 1 displays the spatial distribution of the SURFRAD, AmeriFlux, and BSRN sites.

III. Method

There were two steps to estimate the daily R_n. First, the surface R_{ns} was derived from the satellite TOA observations using an AHM model instead of a traditional LUT. The AHM couples extensive radiative transfer simulations with a random forest (RF) model to ensure retrieval accuracy, and it does not require cloud mask products as constraints. Then, leveraging other auxiliary information, including MERRA-2 Ta and RH and GLASS FVC, the AHM-derived all-sky R_{ns} was extended to estimate R_n. Fig. 2 displays the flowchart of this study for estimating all-sky R_{ns} and R_n.

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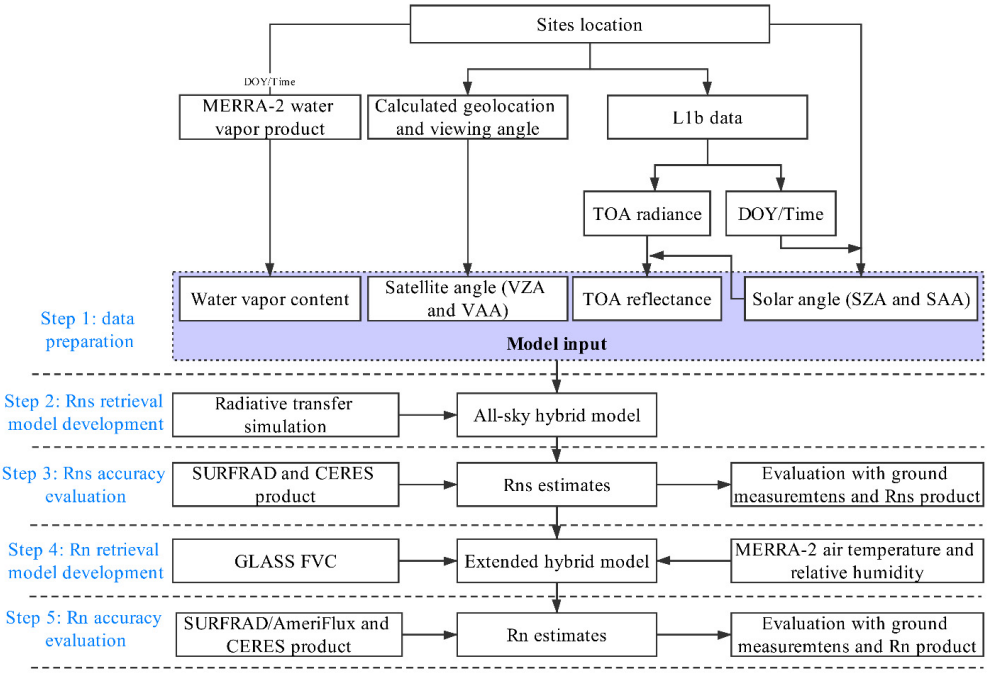


Fig. 2. Flowchart of this study for estimating R_{ns} and R_n.

A. Estimation of Surface R_{ns} Using All-Sky Hybrid Method

Surface R_{ns} linearly links with the upwelling shortwave radiation flux at the TOA [68]. Generally, R_{ns} can be parameterized as follows:

$$R_{ns} = \alpha \times \frac{E_0 \cos(\mathrm{SZA})}{D^2} - \beta F_u \tag{2}$$

where E_0 is the solar radiation at TOA, D is the Earth–sun distance in astronomical unit, and F_u is the TOA USR. Besides, F_u can be modeled by TOA albedo according to

$$F_u = \gamma \times \frac{E_0 \cos(\mathrm{SZA})}{D^2} \tag{3}$$

where γ is the TOA albedo. γ can be estimated from TOA reflectance using the following equation:

$$\gamma = n_0 + \sum n_k \rho_k \tag{4}$$

where ρ_k means TOA reflectance of satellite band k, and n_0 and n_k are the regression coefficients influenced by observation geometry. Thus, considering the above three formulas, R_{ns} can be estimated from satellite-observed TOA reflectance.

To build a reliable and robust relationship between satellite observations and surface R_{ns}, we first performed extensive off-line atmosphere radiative transfer simulations under various surface and atmospheric conditions. When surface and atmospheric variables were set, the radiative transfer model can simulate the corresponding TOA reflectance and surface R_{ns}. Based on the simulation dataset, we can derive a reliable and robust relationship between satellite TOA reflectance and surface R_{ns}. Finally, the relationship was used to real satellite data for estimating surface R_{ns}.

Traditional hybrid models use two LUTs (e.g., linear regressions) to separately estimate clear- and cloudy-sky R_{ns}, which depends on the cloud mask product [1], [24], [56]. Inconsistencies in accuracy often exist between cloud products over different scenes, and accurate cloud property determination typically requires reliable surface reflectance values as an input [57], [58]. Additionally, creating an LUT for real images is time-consuming due to the high-dimensional interpolation operations required [69]. This is particularly relevant to new-generation geostationary satellite data due to the huge data volume [70]. To efficiently estimate R_{ns} without a cloud mask product, an AHM model was used to estimate all-sky R_{ns}. Establishing the AHM followed two steps. First, a comprehensive dataset including TOA reflectance and corresponding surface R_{ns} data was simulated using a radiative transfer model, which can represent different atmospheric conditions and surface features. Second, an empirical model (trained using a machine learning approach) was developed to establish the relationship between TOA reflectance and surface R_{ns} under all-sky conditions.

In this study, the radiative transfer model libRadtran version 2.0.2 [71], [72] was used to perform extensive simulations under representative atmospheric and surface conditions. The open-source libRadtran model was employed as a simulation tool because it exhibits high accuracy and speed [73]–[76]. A total of 245 surface spectra (including various land-cover types) ranging from 0.3–3 μm and at a 1-nm spectral resolution were used to characterize Earth surface features during radiative transfer simulations [77], [78]. Two primary atmospheric parameters were considered for clear-sky conditions: aerosol optical depth (AOD) and water vapor level. In contrast, the cloud optical depth (COD) and water vapor level were the primary parameters considered in the atmospheric radiative transfer simulations under cloudy-sky conditions. Solar-viewing geometries (SZA, VZA, and RAA)

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were also be considered in the simulations. According to previous studies [79]–[81], the simulation parameters were set as follows: SZA (0°–80°, at a 5° interval), VZA (0°–80°, at a 10° interval), RAA (0°–180°, at a 30° interval), AOD (0.01, 0.025, 0.05, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1, 2, and 3), COD (1, 5, 10, 20, 30, 40, 50, 60, 70, 80, and 100), and water vapor level (0, 5, 10, 15, 30, 45, 60, 75, 90, and 105 mm). Previous studies pointed out that different aerosol modes have insignificant impacts on shortwave radiation retrieval for most cases [20], [79], [82]. Based on these previous studies, aerosol types are set to continental. Unlike previous studies [28], [50]–[52], [67], [68], spectral response functions (SRFs) were not directly included in the radiative transfer simulations in this study. We refined and improved the simulation process as follows: 1) hyperspectral TOA reflectance data at a 1-nm spectral resolution was simulated using libRadtran; 2) TOA reflectance of the satellite channels (e.g., GOES-16 and MODIS) were calculated using the simulated hyperspectral TOA reflectance dataset and SRFs based on the following equations; this stage is simple and efficient:

$$\rho_{b(i)} = \frac{\int_{\lambda_{\min}}^{\lambda_{\max}} \mathrm{SRF}(\lambda)\rho(\lambda)d\lambda}{\int_{\lambda_{\min}}^{\lambda_{\max}} \mathrm{SRF}(\lambda)d\lambda} \tag{5}$$

where $\rho_{b(i)}$ is the TOA reflectance of the simulated satellite for the ith band; and $\lambda_{\min}$ and $\lambda_{\max}$ are the minimum and maximum boundaries of the wavelength at the ith band, respectively. SRF(λ) represents the SRF at the ith band. In [24], [54], and [56], radiative transfer simulation should be performed repeatedly when the used satellite is changed, and the process is extremely time-consuming. The simulation refinement and improvement allows the AHM model to be applicable to other satellites for estimating R_{ns} when SRFs are offered.

Machine learning methods have recently attracted greater attention within the remote sensing community due to their outstanding fitting power [83]. Compared to other machine learning technologies, such as the support vector machine and artificial neural network, the RF algorithm exhibits unique advantages including faster application, greater simplicity, and fewer parameter adjustments [84]. The RF is composed of multiple decision trees that are fitted using various randomly divided subsets. Each tree provides an estimate, and the final estimate is averaged from the outputs of multiple trees, contributing to high predictive accuracy and avoiding overfitting [84]. During RF model development, the number of trees was set to 100, and the other parameters were set to the default values. Our test indicated that the RF model performance is similar if the tree number is greater than 100 (Fig. 3); however, the model complexity and training cost significantly increase.

Given an actual satellite observation, the AHM generates a corresponding R_{ns} estimate. The AHM does not need any cloud information inputs. Ying *et al.* [69] preliminarily estimated R_{ns} under all-sky conditions from MODIS using an AHM-like scheme. However, further refinement and improvement has been performed in this study: 1) they considered limited surface and atmosphere conditions during the radiative transfer simulation (e.g., nine surface spectra were used;

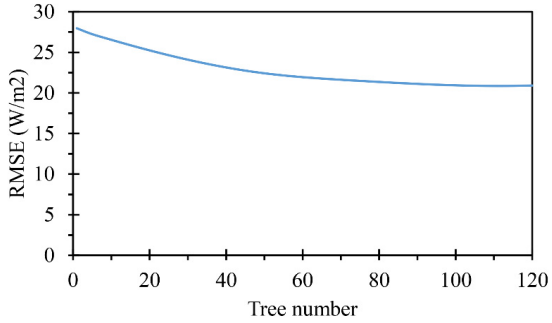


Fig. 3. Determination of the optimal tree number by exploring the relationship between tree number and the modeling RMSE.

default visibility was used as clear-sky optical depth), resulting in that the validation results in their paper are preliminary and the model may be unsuitable for mapping regional R_{ns}; 2) they employed MODIS water vapor product MYD05 to remove the effects of water vapor on R_{ns}, and the failure in R_{ns} retrieval would occur because MYD05 presents data gap under cloudy-sky days; and 3) the ability of spatial R_{ns} maps from AHM-like scheme has not been explored. Because GEOS-16 does not include a band designated for water vapor absorption estimation in the shortwave range, the influence of atmospheric water vapor on R_{ns} was corrected for using the selected water vapor product. Previous studies used a MODIS water vapor product (MOD05/MYD05) to perform water vapor absorption correction [69], [85]. However, spatially continuous R_{ns} distributions cannot be mapped using these data under cloudy-sky conditions due to the missing values in MOD05/MYD05. Thus, this study uses spatio-temporal continuous water vapor levels from the MERRA-2 product to estimate and monitor spatially continuous changes in the surface energy budget. Then, R_{ns} was estimated using the following equation:

$$R_{ns} = f(\mathrm{SZA}, \mathrm{VZA}, \mathrm{RAA}, \omega, \rho_{\mathrm{TOA}}) \tag{6}$$

where f represents the function linking predictors and R_{ns}, which is expressed using an RF model. ω represents the water vapor level. ρ_{TOA} is the TOA reflectance of the four GOES-16 bands (Table I).

The second step in a traditional hybrid model is typically to create two LUTs using a linear regression approach [55], [69], [70]. The LUTs store the specific linear regression coefficients for cases of different solar-viewing geometries and water vapor levels. One LUT is suitable for clear-sky conditions, and the other is used for cloudy-sky conditions. These traditional LUTs were developed to compare the R_{ns} retrieval accuracy from the AHM, particularly under different sky conditions. R_{ns} under different sky conditions was estimated from satellite TOA data under a group of solar-viewing geometry and water vapor level conditions, as expressed by the following equation:

$$R_{ns} = c_0(\mathrm{SZA}, \mathrm{VZA}, \mathrm{RAA}, \omega) + \sum_{i=1}^{b} m_i(SZA, \mathrm{VZA}, \mathrm{RAA}, \omega) \times \rho_{\mathrm{TOA}(i)} \tag{7}$$

where c_0 and m_i are the offset and coefficient of the linear regression for the ith band. The regression coefficients were

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derived from the simulation dataset under a given viewing geometry (i.e., a combination of the SZA, VZA, and RAA) and water vapor concentration. Additionally, two suites of regression coefficients were obtained for different sky conditions (clear or cloudy sky). Because the c_0 and m_i values rely on the sky conditions, a cloud mask product is needed to first distinguish between clear- and cloudy- sky pixels before performing the LUTs. The accuracy of R_{ns} estimates from the traditional LUT method and AHM was evaluated and compared against *in situ* measurements.

B. Extended Hybrid Model for Surface R_n Estimation

The hybrid model cannot be directly employed to estimate all-sky R_n because real-time and accurate cloud parameters are needed to characterize the longwave radiation under cloudy-sky conditions [42]–[44]. Based on the AHM, an extended hybrid model (EHM) was further developed to derive R_n from accurate R_{ns} estimates. The shortwave radiation budget significantly contributes to R_n [45], [48], [65]; thus, R_n can be directly estimated from R_{ns} using their high correlation, which is dependent on the land-cover type and temperature conditions. Previous studies have demonstrated that surface vegetation cover and Ta considerably impact the relationship between R_n and shortwave radiation [48], [50].

The EHM was built to retrieve daily R_n using another RF model; the model inputs include daily R_{ns} estimates, LRD, FVC, daily Ta (including minimum, maximum, and mean values), and daily mean RH. Because the satellite observations are instantaneous, extrapolations of the instantaneous radiation estimation are typically needed to obtain the daily values [24]. Note that daily R_{ns} was integrated and averaged from instantaneous R_{ns} according to the following equation:

$$R_{ns}^{\text{Daily}} = \frac{\int R_{ns}^{\text{Inst}}(t)dt}{T} \tag{8}$$

where $T = 24$ h [24], [56]. Thus, daily R_n was estimated as follows:

$$R_n^{\text{Daily}} = f\left(R_{ns}^{\text{Daily}}, \text{LRD}, \text{FVC}, T_a, \text{RH}\right) \tag{9}$$

f was built using another RF model. To ensure the robustness of the proposed EHM, many ground measurements were collected. Ground measurements from a total of 104 sites from SURFRAD and AmeriFlux networks were used to build and evaluate the EHM. Ten-fold cross-validation (CV) was then performed to evaluate the model performance.

IV. Results

A. Validation of Instantaneous R_{ns} Estimates Derived From LUT and AHM

For R_{ns} estimates derived using the LUT approach, two groups of linear regression coefficients were separately generated using the simulated dataset. One LUT was used for clear-sky pixels and the other for cloudy-sky pixels, requiring that sky conditions must first be distinguished before estimating R_{ns} [41], [55]. Based on the simulation dataset, the impact of correct cloud mask information on the R_{ns} estimation

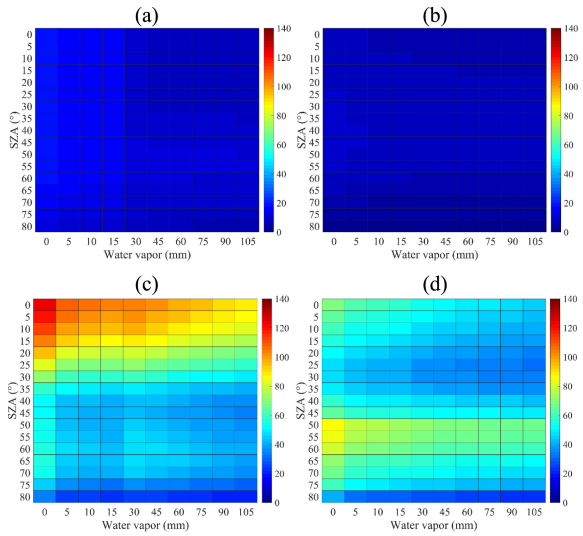


Fig. 4. RMSEs of instantaneous R_{ns} estimates derived from LUT with correct and incorrect cloud information. (a) Using clear-sky LUT to clear-sky pixels, (b) Using cloudy-sky LUT to cloudy-sky pixels, (c) Using clear-sky LUT to cloudy-sky pixels, and (d) Using cloudy-sky LUT to clear-sky pixels.

performance with the LUT method was explored. The clear-sky LUT was used to both clear-sky and cloudy-sky simulation cases for estimating corresponding R_{ns}, and the estimation errors are shown in Fig. 4(a) and (c). Similar analysis was conducted for cloudy-sky LUT, and the results are displayed in Fig. 4(b) and (d). When using the clear-sky LUT for clear-sky pixels, the RMSEs of the estimated R_{ns} range from 6.17 to 18.86 W/m^2. When using the cloudy-sky LUT for cloudy-sky pixels, the RMSEs range from 1.28 to 10.02 W/m^2. In addition, the bias values are near zero because the residual errors of the linear regression are minimized during LUT establishment. In contrast, when using clear-sky LUT for cloudy-sky pixels, the validation results of R_{ns} include RMSEs ranging from 20.19 to 123.15 W/m^2 and biases ranging from 7.5 to 48.74 W/m^2. Furthermore, when using cloudy-sky LUT for clear-sky pixels, the validation results include RMSEs ranging from 22.74 to 85.55 W/m^2 and biases ranging from -30.50 to 1.63 W/m^2.

The accuracy of R_{ns} estimates derived from LUT and AHM approaches with GOES-16 data was evaluated against ground measurements. The validation results under different sky conditions are displayed in Fig. 5. The overall validation results with the LUT method under all-sky conditions exhibit an R^2 of 0.88, an RMSE of 79.72 W/m^2, and a bias of 13.99 W/m^2. The LUT validation R^2, RMSE, and bias under cloudy-sky conditions are 0.83, 92.03 W/m^2, and 25.87 W/m^2, and these values are 0.94, 51.96 W/m^2, and -6.55 W/m^2 under clear-sky conditions, respectively. In contrast, the overall AHM validation results under all-sky conditions display an R^2 of 0.89, an RMSE of 75.28 W/m^2, and a bias of 5.05 W/m^2. The AHM validation R^2, RMSE, and bias under cloudy-sky conditions are 0.84, 85.73 W/m^2, and 13.15 W/m^2, and these values are 0.94, 52.52 W/m^2, and -8.96 W/m^2 for clear-sky conditions, respectively. The results demonstrate that the AHM under cloudy-sky conditions performs better than the traditional LUT approach with a lower RMSE (by 6.30 W/m^2) and bias (by 12.72 W/m^2) and higher R^2 (by 0.01). The accuracy

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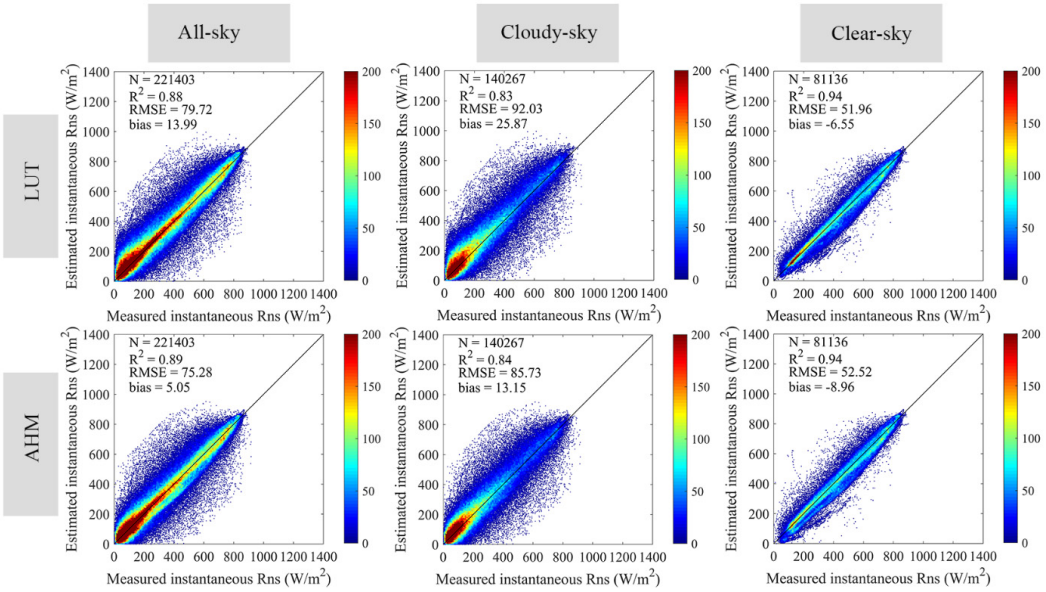


Fig. 5. Validation results of instantaneous R_{ns} estimates under different sky conditions based on the LUT and AHM algorithms. Note that the color bar means samples density.

TABLE II
VALIDATION RESULTS OF LUT- AND AHM-DERIVED INSTANTANEOUS R_{ns} ESTIMATES AT DIFFERENT SITES

Site	R^2	RMSE (W/m^2)	bias (W/m^2)	R^2	RMSE (W/m^2)	bias (W/m^2)
	LUT			AHM		
FPK	0.88	80.53	14.82	0.92	62.85	-4.27
SXF	0.88	73.16	6.29	0.90	67.84	-5.58
PSU	0.87	81.32	29.78	0.87	79.84	24.04
BON	0.91	68.36	23.15	0.91	66.05	17.14
TBL	0.81	97.77	11.42	0.83	94.03	-0.74
DRA	0.86	81.91	-14.46	0.89	76.28	-12.67
GWN	0.91	73.93	25.42	0.91	72.7	24.35
All	0.88	79.72	13.99	0.89	75.28	5.05

of the AHM under clear-sky conditions is comparable to that of the LUT, because the differences in the RMSE and bias are less than 0.6 and 2.5 W/m^2, respectively. The accuracy under the all-sky conditions of the AHM is superior to that of LUT due to the improved performance of the AHM under cloudy-sky conditions.

Table II displays the validation results of the LUT- and AHM-derived instantaneous R_{ns} estimates at different sites. Compared to the LUT, the AHM exhibits improved accuracy at each site. Thus, we used the AHM to estimate the all-sky R_{ns}. When R_{ns} retrieval was performed on a PC Intel Core i7-7700 processor (3.60 GHz) and 32-GB RAM, the LUT method required tens of minutes, while the AHM only required several seconds. With the advent of remote-sensing big data, the computational efficiency of the model should be considered [86], particularly for new-generation geostationary satellites with high spatio-temporal resolutions, which acquire large volumes of data every day [86].

R_{ns} is influenced by both atmospheric conditions and surface features, such as the SZA, clearness index (KT), water vapor, and surface albedo [3]. KT is defined as the proportion of real DSR to extraterrestrial solar radiation, which is a suitable index for indicating the total scattering and absorption process of all atmospheric constitutes [87]. The scatter plots of the AHM validation error with SZA, KT, water vapor, and surface albedo are displayed in Fig. 6. The surface albedo was calculated from *in situ* measurements; albedo = USR/DSR. The results demonstrate that nearly all of the scatter points are close to y = 0 line. The correlation coefficients (R) between the validation errors and these four variables are small and insignificant, ranging from -0.0604 to 0.1723 ($P < 0.01$). These results demonstrate that the developed AHM can sufficiently explain the relationship between R_{ns} and atmospheric and surface parameters with no significant system residual errors.

B. Validation of Daily R_{ns} Estimates

The accuracy of daily R_{ns} estimates from GOES-16 was also evaluated against ground measurements. For comparison, daily R_{ns} retrieval accuracy from MODIS using the AHM estimation scheme was evaluated for the same periods and sites. As a reliable reference product with reasonable accuracy [88], the CERES-SYN product was used to derive the daily R_{ns}, which was validated using the same samples. The scatter plots of the validation results are presented in Fig. 7. The spatial resolution of the estimated R_{ns} from GOES-16 and MODIS is 1 km, but the CERES-SYN exhibits a coarse resolution of 1°. The daily R_{ns} estimates were integrated from temporally discrete satellite observations. There are 5-min interval and 1-h observations within a day for GOES-16 and

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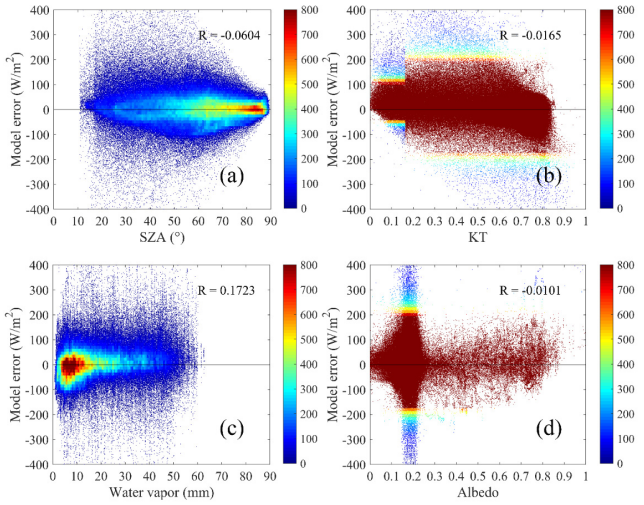


Fig. 6. Scatter plots of the AHM validation error for instantaneous R_{ns} estimates. (a) Model error and SZA. (b) Model error and KT. (c) Model error and water vapor. (d) Model error and surface albedo. Note that the color bar means samples density.

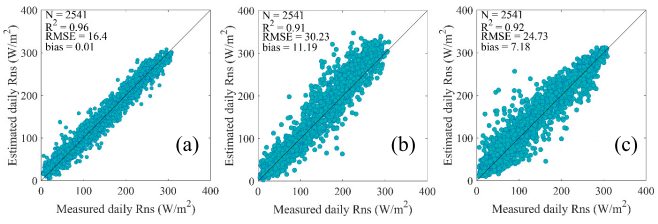


Fig. 7. Validation results of daily R_{ns} estimates derived from (a) GOES-16, (b) MODIS, and (c) CERES-SYN.

CERES-SYN, while MODIS has about two daytime observations (∼10:30 and 13:30 UTC). Of these three daily R_{ns} products, GOES-16 displays the highest accuracy with an R^2 of 0.96, an RMSE of 16.4 W/m^2, and a bias of 0.01 W/m^2 (Fig. 7) due to its high spatio-temporal resolution. In contrast, limited by sparse temporal observations, the MODIS results do not perform well with $R^2 = 0.91$, RMSE = 30.23 W/m^2, and bias = 11.19 W/m^2. The CERES-SYN validation R^2, RMSE, and bias are 0.92, 24.73 W/m^2, and 7.18 W/m^2, respectively. The results demonstrate that CERES-SYN has reasonable accuracy in R_{ns} estimates. It suggests that temporal resolution may be a vital parameter contributing to the daily R_{ns} estimate accuracy. This is likely why some remote sensing products with high temporal resolutions exhibit a coarse spatial resolution but acceptable daily estimates.

Table III displays the validation results of daily R_{ns} estimates derived from GOES-16, MODIS, and CERES at different sites. The results indicate that GEOS-16 exhibits the highest accuracy, followed by CERES and MODIS. To examine the effect of temporal resolution on the daily R_{ns} estimation accuracy, instantaneous R_{ns} estimates at different temporal resolutions (i.e., 5 min, 15 min, half h, h, 1.5 h, and 3 h) were integrated into the daily values. The validation results at different temporal resolutions are displayed in Fig. 8. The RMSEs range from 16.41 to 23.32 W/m^2, and biases range from −3.98 to 0.01 W/m^2. The RMSE and bias decrease

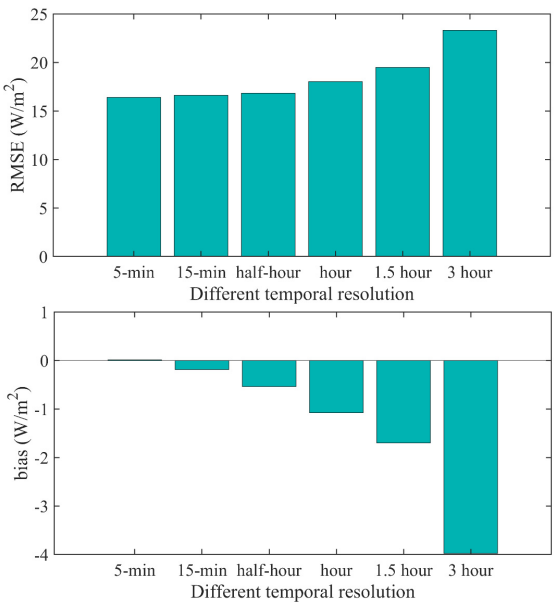


Fig. 8. Validation RMSE and bias results of GOES-16 estimated daily R_{ns} when instantaneous estimates at different temporal resolutions are integrated to daily values.

with an increase in the temporal resolution, demonstrating the importance of high observation frequency for accurate daily R_{ns} estimation. This is probably because dense diurnal observations can better capture and model random cloud occurrences.

C. Mapping Daily R_{ns} Over CONUS

As illustrated in Fig. 9, the AHM was further used with actual GOES-16 images to map the daily 1-km R_{ns} over the CONUS on April 2, July 16, October 30, and December 29, 2018. These four days were randomly selected to demonstrate the ability of AHM to map R_{ns} under different seasons and using GOES-16 data. The cloud occurrence frequency presents significant spatio-temporal variations in R_{ns} over the CONUS. To assess the overall spatial patterns of R_{ns} maps from GOES-16, the daily R_{ns} estimates derived from the CERES-SYN product with 1° spatial resolution for the same day are also displayed in Fig. 9. The R_{ns} estimates from GOES-16 and CERES-SYN exhibit similar spatial distribution patterns in the four seasons.

To investigate the differences between GOES-16- and CERES-derived daily R_{ns} estimates, GOES-16 data was aggregated to match the spatial resolution of the CERES data for comparison. Scatter plots and statistics between these two types of R_{ns} products on four days are displayed in Fig. 10. There are significant relationships and high consistency between GOES-16 and CERES. The R^2 ranges from 0.86 to 0.95; the overall RMSEs are small, within the range of 10.47 to 15.68 W/m^2; and the bias ranges from 0.72 to 6.55 W/m^2. According to Figs. 9 and 10, there is high consistency between the GOES-16 and CERES product estimates, which indicates that the accuracy of CERES is reasonable with its high temporal resolution of 1 hour. The

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TABLE III
VALIDATION RESULTS OF GOES-16, MODIS, AND CERES-DERIVED DAILY R_{ns} ESTIMATES AT SURFRAD SITES

Site	R^2	RMSE (W/m^2)	bias (W/m^2)	R^2	RMSE (W/m^2)	bias (W/m^2)	R^2	RMSE (W/m^2)	bias (W/m^2)
	GOES-16			MODIS			CERES		
FPK	0.98	14.22	-1.58	0.94	30.39	14.11	0.95	22.01	2.16
SXF	0.96	15.83	-3.9	0.91	31.72	6.29	0.92	24.84	-2.79
PSU	0.96	16.94	7.47	0.92	32.18	18.95	0.93	25.49	14.83
BON	0.98	13.38	4.68	0.94	27.86	13.83	0.90	30.10	13.79
TBL	0.93	20.36	-1.84	0.86	35.25	10.52	0.81	36.17	13.22
DRA	0.96	18.65	-12.56	0.92	22.69	-2.31	0.95	16.39	-1.48
GWN	0.98	14.37	7.86	0.88	34.27	19.21	0.92	26.76	14.91
All	0.96	16.4	0.01	0.91	30.23	11.19	0.92	24.73	7.18

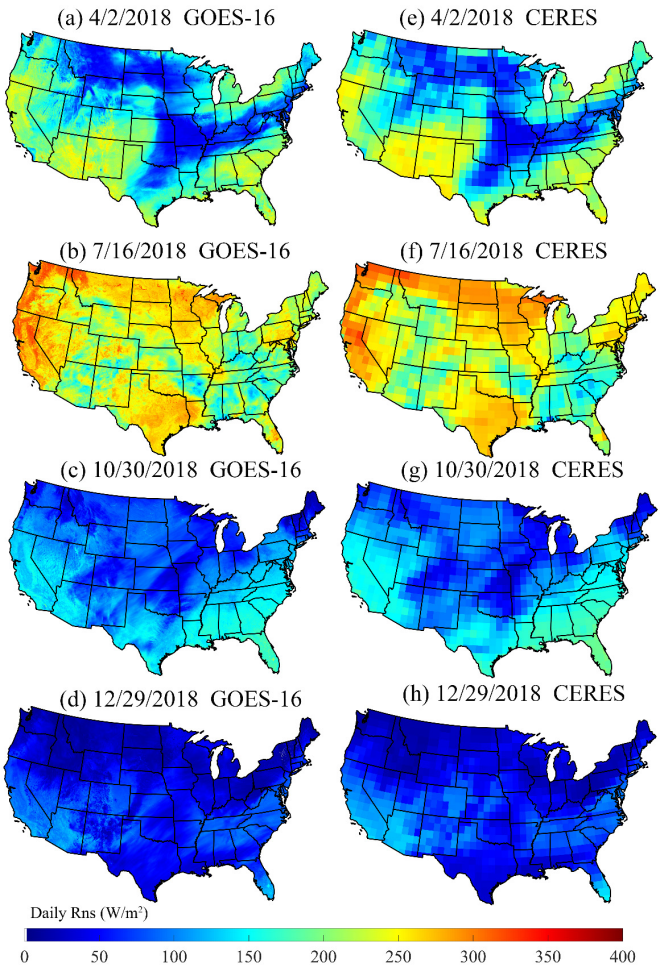


Fig. 9. Mapping of daily R_{ns} over the CONUS on April 2, July 16, October 30, and December 29, 2018. (a),–(d) R_{ns} estimates derived from GOES-16 data at 1 km spatial resolution. (e)–(h) R_{ns} estimates derived from CERES-SYN product at 1° spatial resolution.

spatial intercomparison between these two products further demonstrates that temporal resolution is a critical parameter contributing to the accuracy of the daily R_{ns} estimates.

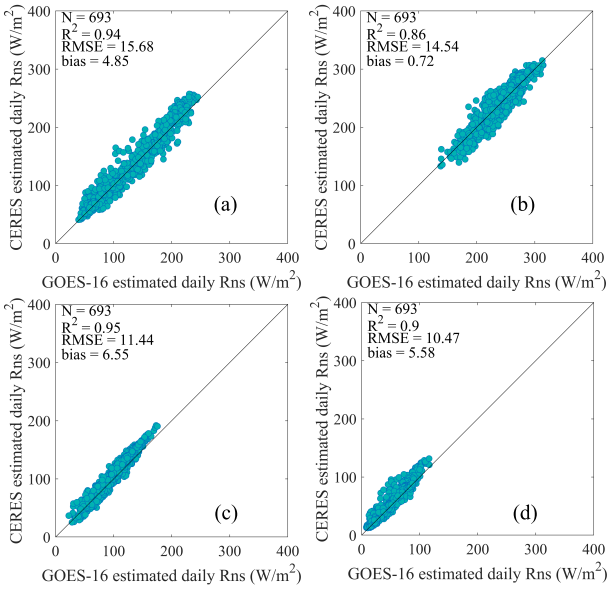


Fig. 10. Scatter plot between daily R_{ns} estimates on April 2, July 16, October 30, and December 29, 2018 derived from GOES-16 and CERES data using matched pairs from the maps of Fig. 9. GOES-16 pixels are aggregated to the same spatial resolution of CERES for the comparison. (a) 4/2/2018, (b) 7/16/2018, (c) 10/30/2018, and (d) 12/29/2018.

D. Evaluation of EHM-Derived Daily R_n Estimation

The EHM was developed to derive the daily R_n from R_{ns} estimates. The LRD was used to consider diurnal changes in shortwave and longwave radiation across different seasons. The ground-measured samples collected in this study exhibited an LRD ranging from 0.34 to 0.67. To investigate the impact of the LRD on the relationship between R_{ns} and R_n, the dataset was divided into four groups in increments of 0.1. Table IV displays the statistics of the linear model established between the daily R_{ns} and R_n. The R^2 and RMSE values change with LRD because R_{ns} under a larger LRD contributes more to the daily R_n. R^2 presents an increasing trend, and the RMSE displays an inverse trend with an increasing LRD. The relationship built using all samples exhibits an R^2 of 0.79 and RMSE of 31.68 W/m^2. Additionally, the linear regression

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TABLE IV
STATISTICS FOR LINEAR MODELS BUILT BETWEEN DAILY R_{ns} AND R_n UNDER DIFFERENT LRD VALUES

Min LRD	Max LRD	$R_n = a \times R_{ns} + b$		R^2	RMSE (W/m^2)
		a	b		
0.34	0.44	0.45	-8.01	0.35	23.62
0.44	0.54	0.69	-10.18	0.64	30.01
0.54	0.64	0.63	9.35	0.57	34.81
0.64	0.67	0.76	-8.79	0.81	22.44
All		0.74	-19.36	0.79	31.68

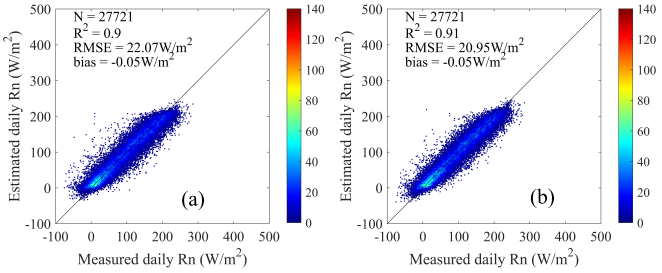


Fig. 11. Scatter plots of ten-fold cross-validation results for R_n estimates using the EHM models. (a) The EHM without LRD and (b) The EHM with LRD.

coefficients are different, indicating that the relationship between R_{ns} and R_n is nonlinear and complex. The RF model was used in this study to solve this nonlinear regression issue.

Fig. 11 displays the CV results with and without the LRD as a model input. The EHM without the LRD performs well; the CV results exhibit an R^2 of 0.90 and an RMSE of 22.07 W/m^2. In Table IV, the modeled R_n via the simple linear regression exhibits an RMSE of 31.68 W/m^2. However, R_n modeled using the EHM exhibits a lower RMSE ($\sim$17.58 W/m^2). The primary reason for the high R_n retrieval accuracy is that an RF model coupled with accurate R_{ns} estimates and other predictors was employed. Furthermore, the LRD influenced the relationship between R_{ns} and R_n (Table IV). When the LRD was incorporated into the EHM, the CV R^2 and RMSE improved to 0.91 and 20.95 W/m^2 (a decrease in the uncertainty of $\sim$5.07%), respectively. For comparison, the accuracy of CERES-SYN-derived daily R_n was also evaluated using the same samples. According to Fig. 12, the validation R^2, RMSE, and bias are 0.84, 27.83 W/m^2, and 1.6 W/m^2, respectively. Fig. 13 shows the independent validation results of the EHM model using BSRN ground measurements, which is comparable to the ten-fold CV accuracy. The results indicate that the proposed EHM model with LRD is a feasible model for deriving all-sky R_n from the AHM-estimated R_{ns}. The estimated R_n from GOES-16 demonstrates higher accuracy due to the much higher spatial resolution.

E. Mapping R_n Over CONUS

As illustrated in Fig. 14, the carefully evaluated EHM was used to map the daily 1-km R_n over the CONUS on April 2, July 16, October 30, and December 29, 2018. To examine

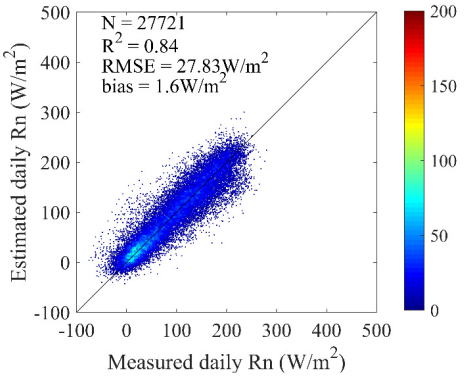


Fig. 12. Scatter plot between R_n estimates derived from CERES-SYN and ground measurements. Note that the y-axis indicates CERES-SYN estimated R_n.

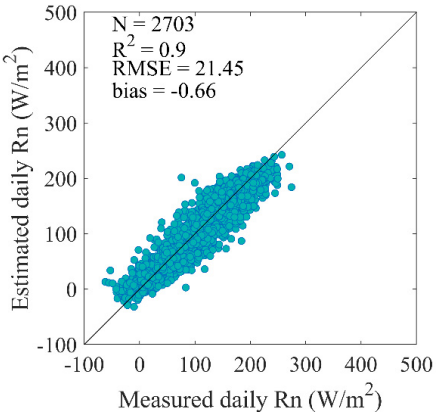


Fig. 13. Evaluation of the EHM model performance against BSRN measurements.

the overall spatial patterns of R_n maps from GOES-16, the daily R_n estimates derived from the CERES-SYN product with 1° spatial resolution for the same day are also displayed in Fig. 14. The results demonstrate that the R_{ns} estimates from GOES-16 and CERES-SYN products exhibit similar spatial distribution patterns in the four seasons. Greater detail can be observed in the R_n maps from GOES-16 than the CERES-SYN maps because of the improved spatial resolution of 1 km. Additionally, the R_{ns} and R_n maps display the same spatial patterns, as illustrated in Figs. 9 and 14. The daily R_n maps indicate that the proposed EHM poses great potential for accurate daily 1-km R_n estimation at the regional scale.

To investigate the differences between GOES-16 and CERES-derived daily R_n estimates, GOES-16 data was aggregated to the same spatial resolution of CERES. Scatter plots and statistics between these two types of R_n products for the four seasons are displayed in Fig. 15. There are significant relationships and high consistency between GOES-16- and CERES-derived daily R_n. The R^2 ranges from 0.41 to 0.82; the RMSEs are small, within the range of 13.08 to 18.04 W/m^2; and the bias ranges from -5.3 to 7.13 W/m^2.

V. DISCUSSION

The results demonstrate that if correct cloud information is used in the LUT, the estimated R_{ns} is sufficiently accurate.

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TABLE V

SAMPLE OF SPECIFIC COEFFICIENTS OF LUT (VZA = 40°, RAA = 30°, AND WATER VAPOR = 15 mm) WITH DIFFERENT SZAS. NOTE THAT DATA LOCATED BEFORE AND AFTER THE OBLIQUE LINE ARE COEFFICIENTS UNDER CLEAR-SKY AND CLOUDY-SKY CONDITIONS, RESPECTIVELY

SZA (°)	c_0	m_1	m_2	m_3	m_5
0	1113.75/1128.85	-395.49/-370.28	-215.92/-238.48	-347.68/-335.35	-181.91/-181.65
5	1108.23/1105.98	-394.59/-389.96	-214.67/-221.73	-346.25/-331.67	-181.30/-187.15
10	1093.53/1080.49	-391.32/-399.11	-211.18/-207.47	-341.92/-325.35	-179.31/-190.90
15	1069.86/1051.89	-385.72/-397.19	-205.48/-198.00	-334.73/-317.79	-175.96/-189.13
20	1037.45/1017.11	-377.84/-386.52	-197.61/-191.89	-324.74/-308.80	-171.28/-183.02
25	996.63/974.50	-367.80/-368.87	-187.61/-187.86	-312.05/-298.07	-165.33/-173.76
30	947.80/924.09	-355.76/-344.83	-175.53/-185.51	-296.75/-285.64	-158.18/-161.53
35	891.43/866.85	-341.84/-315.17	-161.47/-184.26	-278.98/-271.66	-149.90/-146.20
40	828.14/803.46	-326.20/-281.94	-145.54/-182.47	-258.89/-256.02	-140.58/-128.28
45	758.52/734.00	-308.94/-248.16	-127.95/-177.69	-236.70/-238.40	-130.29/-108.93
50	683.32/658.98	-290.11/-216.47	-108.97/-167.79	-212.62/-218.56	-119.11/-89.25
55	603.34/579.26	-269.46/-188.85	-89.17/-151.28	-186.95/-196.37	-107.07/-70.25
60	519.44/495.90	-246.42/-166.40	-69.41/-127.67	-160.05/-171.77	-94.12/-53.25
65	432.54/410.22	-219.79/-147.79	-51.06/-98.86	-132.38/-144.90	-80.16/-39.71
70	343.50/323.70	-187.26/-128.41	-36.38/-69.88	-104.52/-116.35	-64.92/-30.22
75	252.86/237.49	-143.68/-102.49	-29.87/-46.81	-77.38/-87.06	-47.88/-23.98
80	159.01/152.03	-73.35/-68.59	-43.66/-31.29	-52.66/-57.77	-27.50/-18.67

The accuracy of the estimated R_{ns} is sensitive to cloud information, and the estimation errors substantially increase if incorrect cloud information is incorporated into the LUT. This is because the stored coefficients in the LUT are different for clear- and cloudy-sky conditions, and extrapolation from the clear-sky LUT to the cloudy-sky LUT or vice versa does not lead to a reliable estimation. Given set values for the other parameters (VZA = 40°, RAA = 30°, and water vapor = 15 mm), Table V shows that the LUT coefficients under clear- and cloudy-sky conditions are different with varying SZA and thus misuse of LUT can introduce additional errors. Incorrect cloudy-sky pixels typically introduce larger errors than false clear-sky pixels. The errors induced by incorrect cloud mask information depend on the SZA and water vapor levels, according to Fig. 3(c) and (d). The RMSEs decrease with increasing water vapor, which may be due to the lower absolute R_{ns} levels for cases with high water vapor values (e.g., 30 mm). The RMSEs typically first decrease and then increase with an increasing SZA. This may be due to the fact that the differences in LUT's coefficients decrease and then increase with an increasing SZA. For R_{ns} estimates derived from the AHM approach, the primary advantage is that no cloud mask product is needed. This means that AHM simplifies the retrieval process, and the accuracy of the R_{ns} estimates is not sensitive to a cloud mask product. This trait may carry great implications for generating R_{ns} using data from multiple satellites because cloud products are typically inconsistent between them.

A previous study illustrated that cloud types present relatively minor impacts on shortwave radiation retrieval [56]. It is usually difficult to derive optical properties of ice clouds, and ice crystal scattering models have large uncertainties [89]. Generally, it is a feasible scheme that clouds are assumed to be water clouds [20]. Thus, although water cloud was set as cloud type, the AHM model can still work well and estimate R_{ns} without significant bias, which is consistent with [20], [79], [80], and [82].

Greater detail can be observed in the R_{ns} maps from GOES-16 than in the CERES-SYN product due to its higher spatial resolution. For example, R_{ns} levels in small-scale rivers and inland lakes are higher than their surroundings due to their low surface albedo values. Some small weather disturbances, such as the presence of broken or thin clouds, are reflected in the 1-km R_{ns} maps. Maps of the daily R_{ns} indicate that our developed AHM poses great potential for estimating spatio-temporally continuous daily 1-km R_{ns} over the CONUS.

Compared to previous studies that used polar-orbiting satellite data to estimate R_n (RMSEs = 31.61–38 W/m^2) [10], [24], [30], [37], the EHM-estimated R_n exhibited superior performance. The results demonstrate that: 1) improved R_{ns} estimation contributes to R_n retrieval with high accuracy; 2) the impact of the LRD should be considered during R_n retrieval; and 3) incorporating the LRD into the machine learning approach is feasible.

The proposed approach may introduce errors for estimating R_{ns}/R_n over mountain areas, because topographic effects are not considered within the AHM and EHM models. Topographic effects usually change with the spatial resolution of pixels [90]. Given the assumption of flat surface, estimation of R_{ns}/R_n at spatial resolution of 1 km will generate uncertainties about 3.39%–6.17% [91]. Although this study estimated 1-km R_{ns}/R_n has satisfactory performance without significant bias,

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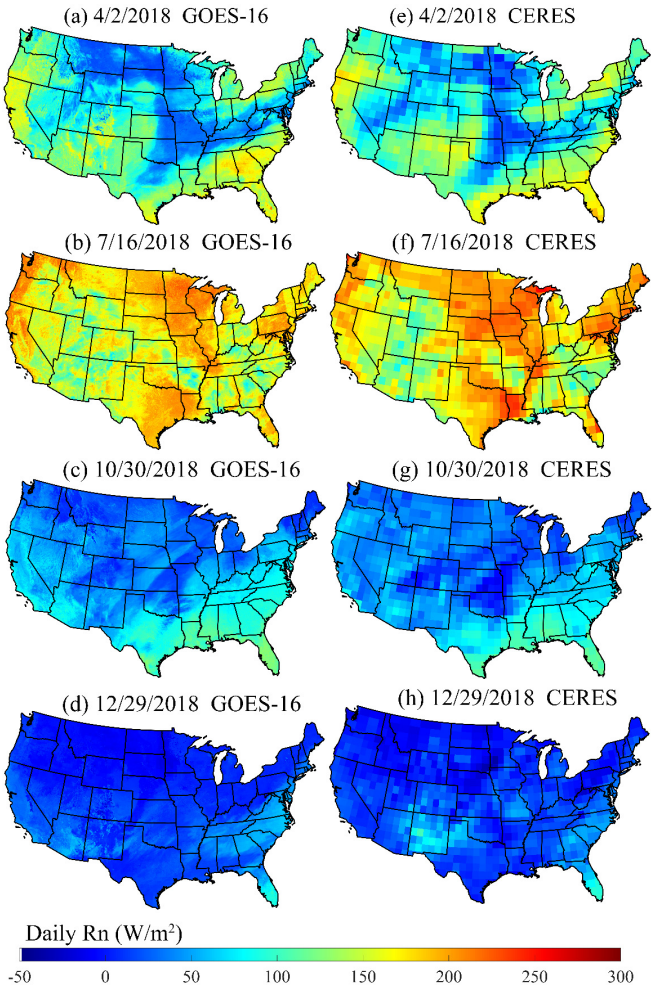


Fig. 14. Mapping of daily R_n over the CONUS on April 2, July 16, October 30, and December 29, 2018. (a)–(d) are R_n estimates derived from GOES-16 data at 1-km spatial resolution. (e)–(h) R_n estimates derived from CERES-SYN product at 1° spatial resolution.

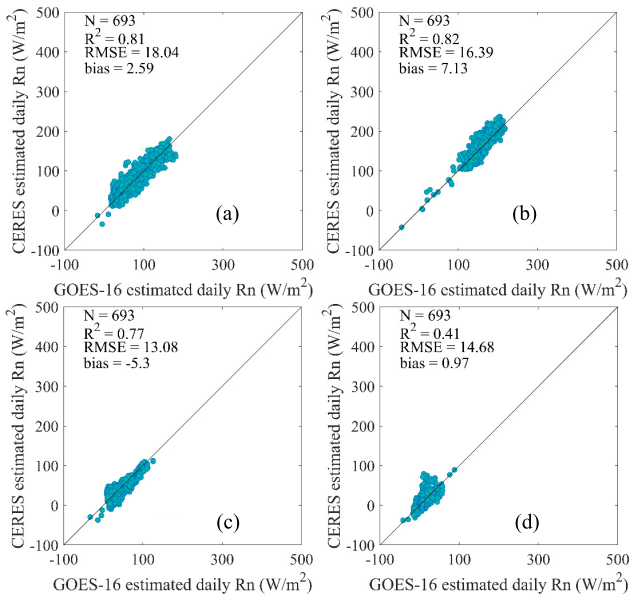


Fig. 15. Scatter plots between daily R_n estimates on April 2, July 16, October 30, and December 29, 2018 derived from GOES-16 and CERES data using matched pairs from the maps of Fig. 14. GOES-16 pixels are aggregated to the same spatial resolution of CERES for the comparison. (a) 4/2/2018, (b) 7/16/2018, (c) 10/30/2018, and (d) 12/29/2018.

the retrieval accuracy may be further improved over mountain areas by considering topographic effects.

The improved satellite observation frequency contributes to the higher retrieval accuracy of R_{ns} estimation. Furthermore, accurate R_{ns} estimates contribute to the more accurate R_n estimation. This study combines the advantages of radiative transfer modeling and machine learning algorithms to improve surface energy budget monitoring via multispectral and multitemporal geostationary remote sensing data. Previous studies needed to perform radiative transfer simulation again when the employed data source changed to another satellite [24], [54], [56], [85]. In this study, the hyperspectral TOA reflectance dataset at a 1-nm spectral resolution was simulated once using libRadtran. Thus, the proposed AHM can be easily applied to other new-generation satellite data, such as Himawari-8, MTG, and Fenyun-4 data, and only two steps should be recalculated: 1) TOA reflectance of a specific satellite can be recomputed by coupling the simulated hyperspectral TOA reflectance dataset with SRFs and 2) an RF model suited to the new satellite should be recalibrated. The AHM is an all-sky approach and independent of cloud products. This advantage may help to generate consistent R_{ns} estimates from these four geostationary satellites. On the other hand, the EHM is independent of a specific satellite, and model parameter recalibration may not be required. Overall, this study presents a feasible and transferable scheme to generate accurate high-spatio-temporal-resolution R_{ns}/R_n estimates by combining multiple new-generation geostationary satellites. However, geostationary satellites do not have spatial coverage at high latitudes. Polar-orbiting MODIS contains many more observations at high latitudes (∼6–20 observations/d) than mid–low latitudes (∼2 observations/d). Our previous work has used multispectral and multitemporal observations from MODIS to estimate daily all-sky 1-km R_n at high latitudes [17]. Thus, it is possible to generate a global daily all-sky 1-km R_n dataset in the future.

VI. Conclusion

This study evaluated the potential of new-generation geostationary satellite GOES-16 data for estimating R_{ns} and R_n. First, an AHM model was proposed to estimate R_{ns} under all-sky conditions. Compared to the traditional LUT approach, the AHM was independent on a cloud mask product; thus, its performance was less affected by cloud contamination than the traditional approach. In addition, the computational efficiency of AHM is superior to that of the LUT method, which is significant for the real-time processing of large volumes of geostationary satellite data, particularly for GOES-16 data which exhibit high spatio-temporal resolutions. The AHM validation errors demonstrated an insignificant relationship

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with the atmospheric and surface parameters, indicating that the AHM accurately estimated all-sky R_{ns} with no significant system residual errors. In sum, the proposed AHM not only simplifies and improves the efficiency of the R_{ns} retrieval process compared to the traditional LUT approach but also presents higher accuracy. Thus, AHM is recommended to estimate R_{ns} under all-sky conditions. Furthermore, the refinement and improvement in the radiative transfer simulation allow the AHM to be easily applied to any multispectral satellite datasets without repeated simulation. The results demonstrate that GOES-16-derived daily 1-km R_{ns} is more accurate than the 1-km MODIS and 1° CERES-SYN products due to its high resolution. The daily R_{ns} validation RMSE and bias decrease with an increase in the temporal resolution, indicating that accurate daily R_{ns} estimation needs multitemporal and multiple daily observations. Dense diurnal observations can capture random occurrences of clouds. The daily 1-km R_{ns} maps from GOES-16 over the CONUS exhibit similar spatial distribution patterns and high consistency with those of CERES-SYN products.

An EHM used to estimate R_n from R_{ns} was developed, avoiding the use of the clear-sky albedo dataset. The results suggest that the LRD should be considered in the R_n retrieval model because it significantly impacts the relationship between R_{ns} and R_n. The CV results of the EHM with the LRD included an R^2 of 0.91 and an RMSE of 20.95 W/m^2. The validation results of GOES-16-derived R_n estimates exhibited improved accuracy compared to those of the CERES-SYN product ($R^2 = 0.84$, RMSE $= 27.83$ W/m^2, and bias $= 1.6$ W/m^2) and previous studies. This is primarily because an RF model coupling accurate 1-km R_{ns} estimates with other predictors was employed. Maps of daily 1-km R_n over the CONUS in different seasons also demonstrate similar spatial patterns to those of the CERES-SYN product. The results indicate that the proposed EHM is a feasible method to generate accurate daily all-sky 1-km R_n estimates. This study offers an efficient scheme to improve R_n retrieval accuracy by taking the advantages of radiative transfer simulation, machine learning, and multispectral and multitemporal satellite observations. In addition, the proposed model can be easily adapted to data from other new-generation geostationary satellites, such as Himawari-8, MTG, and Fenyun-4 data, providing great potential to produce daily 1-km R_{ns}/R_n products at mid–low latitudes, provided more validation efforts regarding the statistical relationship between R_{ns} and R_n. Combined with our earlier work on using MODIS data to estimate R_n over high latitudes [17], it provides a feasible and reliable approach for mapping global daily all-wave all-sky surface net radiation with high spatial and temporal resolutions.

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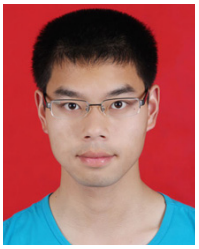
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